

COMPREHENSIVE BRAIN CT ANALYSIS SYSTEM PORTAL

Automated Detection & Classification of Intracranial Hemorrhages with Integrated Clinical Risk Assessment

I. PATIENT & CLINICAL METADATA

Patient Name:	Test Patient	Patient Email:	test@patient.com
Report ID:	BHD-#000123	Analysis Date:	2026-06-17 11:13:26 UTC
Imaging Mode:	Brain CT / MRI Scan	System Core:	PyTorch CPU-Inference

II. CLINICAL ASSESSMENT & FINDINGS

WARNING: ACUTE BRAIN HEMORRHAGE DETECTED

Diagnostic Index	Estimated Value	Clinical Interpretation
Inference Confidence	99.5%	AI confidence of classification based on deep network features.
Hemorrhage Detection Score	99.5%	Estimated likelihood/presence score of brain hemorrhage.
Hemorrhage Severity Index	7.8%	Cross-sectional percentage of brain tissue affected by bleeding.
Stroke Risk Probability	42.51824%	Patient presents a Moderate stroke risk level.
Epilepsy Risk Probability	59.1%	Post-hemorrhage seizure occurrence risk assessment.
Intervention Delay Index (IDI)	0.0000	Urgency index based on hemorrhage type probabilities and presence.
Recommended Care Timeline	Routine (>24 hours)	Immediate, Urgent, Semi-Urgent, or Routine action recommendation.
Hematoma Expansion Rate (HER)	0.00%	Predicted risk of rapid bleeding volume expansion.
Seizure Risk Score (SRS)	0.0000	Estimates long-term seizure risk based on proximity and hemorrhage type.
Hemorrhage Location	Subdural Hematoma	Confidence: 85% in anatomical classification.
Model Accuracy	99.5%	Dataset: real-time

III. EMERGENCY FIRST-AID RECOMMENDATIONS

[EMERGENCY] EMERGENCY ALERT - Hemorrhage Detected

IMMEDIATE ACTIONS:

- Call emergency services (911/999) immediately
- Position patient in recovery position (on side)
- Monitor vital signs (BP, pulse, respiration)
- Keep airway clear, place in recovery position

MODERATE-RISK PROTOCOL:

- Urgent neurology consultation within 30 minutes
- Prepare for possible ICU admission
- Monitor for symptom progression
- Establish IV access for emergency medications

SUBDURAL HEMATOMA PROTOCOL:

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Disclaimer: Automated assessment using deep feature extraction. Always correlate findings with patient symptoms and expert radiological consensus.

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- Monitor for gradual consciousness decline
- Manage intracranial pressure (ICP)
- Assess motor response and hemiparesis

GENERAL MEASURES:

- NPO (nothing by mouth) until airway secured
- Head elevation 30 degrees (if possible)
- Avoid sudden movements
- Keep room quiet and dark
- Document exact time of symptom onset

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IV. SCANS & GRAD-CAM VISUALIZATIONS

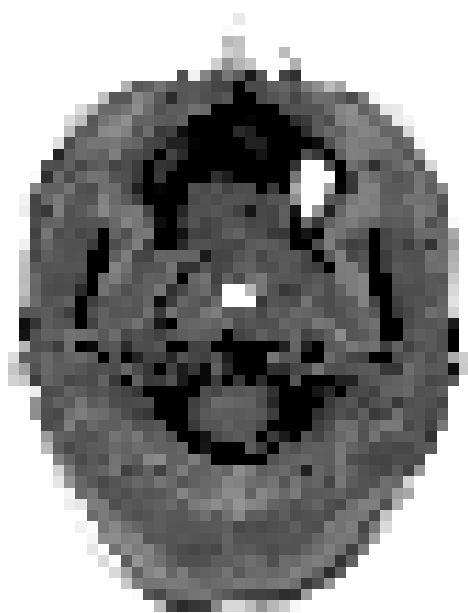


Figure A: Original CT/MRI Scan Input

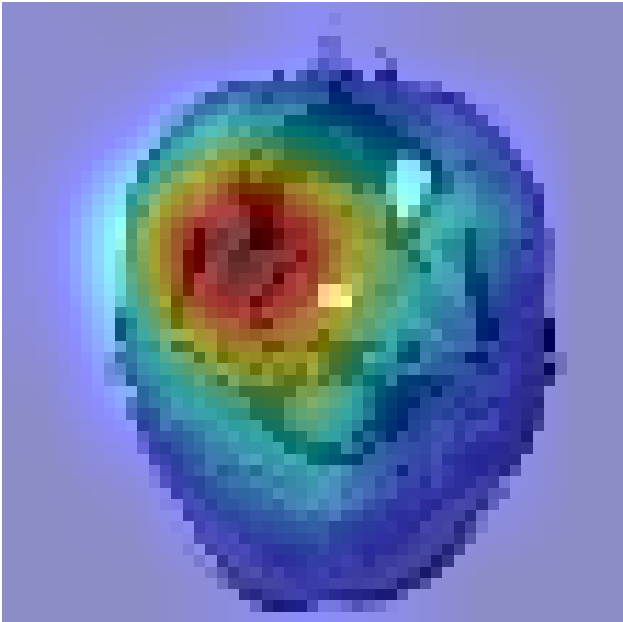


Figure B: Hybrid Grad-CAM Localization Map

Visualization Guide: The Grad-CAM heatmap highlights active spatial regions driving the model prediction. Red/hot-colored focal spots indicate highly suspicious tissue density zones corresponding to hyperdense hematoma accumulations. Green/blue signals show low-probability texture contexts.